Investigating Phase Limitations in STFT-Based Speech Enhancement

ELEC5305 Project

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GitHub project site · GitHub repository · Demonstration video

Abstract

Classical short-time Fourier transform (STFT) enhancement often reconstructs an estimated speech magnitude with the original mixture phase. This project asks how much that choice limits performance across noise and input signal-to-noise ratio (SNR) conditions when the estimated magnitude is fixed. Spectral subtraction and Wiener filtering were evaluated on 360 mixtures from 40 VoiceBank utterances at three noise types and three SNRs. For spectral subtraction, substituting clean reference phase added 0.0170 STOI, 0.0661 PESQ and 1.96 dB SI-SDR on average. STOI improved on all 360 pairs, while some PESQ and SI-SDR scores decreased. After subtracting the mean within each noise-by-SNR cell, the association between phase disagreement and STOI gain remained positive ($r = 0.643$). The corresponding PESQ association weakened from -0.523 to -0.068. The clean-magnitude diagnostic showed further headroom with mixture phase retained. On 167 matched mixtures, pretrained MP-SENet improved PESQ over clean-phase spectral subtraction in every pair. The controlled phase swap establishes a measurable, metric-dependent limitation of phase reuse under the tested conditions. A fixed-magnitude MP-SENet ablation on the 167 mixtures gave mean gains of 0.0040 STOI, 0.1176 PESQ and 1.59 dB SI-SDR. Mean PESQ fell slightly in all three white-noise conditions.

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1 Introduction

1.1 Project overview

Background noise can obscure speech in single-channel recordings and reduce its intelligibility and quality. This project examines a common choice in short-time Fourier transform (STFT) enhancement. Spectral subtraction and Wiener filtering estimate a cleaner magnitude spectrum and typically reconstruct with the original mixture phase. The experiment changes the phase at synthesis while keeping each estimated magnitude fixed.

1.2 Background and motivation

Comparing two complete enhancement algorithms cannot isolate the contribution of phase because their magnitude estimates also differ. A paired phase swap turns each estimator into its own control. Clean reference phase is available in constructed mixtures, allowing a direct diagnostic of noisy-phase reuse. The literature review explains how this control relates to earlier phase research and to systems that estimate complex spectra.

1.3 Research question

How much does reusing noisy phase limit classical STFT speech enhancement across noise conditions and input SNRs when the estimated magnitude is held fixed?

1.4 Methodology overview

The study uses a Python workflow to combine VoiceBank speech with DEMAND cafeteria and traffic recordings and generated white noise at 0, 5 and 10 dB. The classical experiment compares noisy and clean reference phase at the same estimated magnitude. A clean-magnitude diagnostic keeps noisy phase, and pretrained MP-SENet provides a system comparison on matched inputs. Known noise statistics give the classical estimators a controlled baseline. Section 3 describes the implementation and scoring.

1.4.1 Development after feedback

The initial proposal compared the two classical estimators and planned to report output SNR, SNR improvement and STOI. The revised study kept the estimators and focused on phase reuse. It added the clean-phase and clean-magnitude controls, wrapped phase disagreement, PESQ and SI-SDR. The project contribution is the matched phase-swap design, the noise-by-SNR analysis and the investigation of score decreases. The pretrained network supplies a benchmark.

1.5 Expected outcomes

The project was designed to deliver a reproducible Python implementation, a quantitative estimate of phase-related gains and their variation, and an audio example for qualitative inspection. STOI, PESQ and SI-SDR provide complementary views of intelligibility, quality and waveform fidelity. The repository provides run instructions and archived results, and the video demonstrates the code and audio comparison. Sections 4 to 6 assess what the experiments established and where the evidence remains limited.

2 Literature review

2.1 Magnitude-domain estimators

Boll [1] estimates the noise spectrum from nonspeech intervals. Here, spectral subtraction uses the known noise component and applies a floor to limit excessive attenuation. Wiener filtering uses a speech-to-total power estimate to form a gain [2]. Both expose a magnitude estimate for the paired phase swap. Their use of known noise statistics is a controlled simplification of practical enhancement.

2.2 Oracle reconstruction and phase sensitivity

Paliwal et al. [3] found that the contribution of phase depends on the analysis and reconstruction setup. Their oracle comparisons motivate changing phase at synthesis while holding magnitude fixed. Clean speech is available here through the mixture construction, allowing a diagnostic intervention. The measured effect is conditional on the magnitude supplied to the inverse transform.

Wrapped phase is circular: angles near opposite ends of the interval from minus pi to pi may be close on the circle. A wrapped absolute difference avoids treating this boundary as a large disagreement. Weighting by clean magnitude gives speech-bearing bins greater influence. A single weighted mean still discards where errors occur in time and frequency, which matters when the metrics weight speech structure differently.

2.3 Complex and phase-aware models

Complex ratio masking [4] allows a model to rotate and rescale STFT coefficients. DCCRN [5] estimates a complex representation using convolutional and recurrent layers. MP-SENet [6] estimates magnitude and wrapped phase in parallel. Its 2025 journal extension describes the phase decoder and joint training losses in detail [12]. Parvathala et al. [13] modified the MP-SENet encoder using bispectral features and evaluated both enhancement scores and computation. This extends the comparison to the cost of extracting phase-sensitive features.

2.4 Method selection

Classical estimators expose the magnitude and permit a paired phase swap through one inverse-STFT path. Their access to known noise statistics limits how directly the absolute scores apply to blind enhancement. MP-SENet can operate on the mixture and provides a practical comparison on shared inputs.

2.5 Evaluation choices

Taal et al. [9] motivate STOI through short-time speech-envelope comparisons in frequency bands. PESQ compares reference and degraded audio using a perceptual model [10]. This project uses the wideband extension [14]. Le Roux et al. [11] define SI-SDR through projection onto the clean waveform. Those complementary targets motivate separate reporting. Even with an exact input magnitude, imposing a different phase and applying inverse STFT can change the waveform and the magnitude obtained if that waveform is analysed again.

3 Methods

3.1 Data and STFT settings

The archive contains 40 VoiceBank utterances from speakers p232 and p257 [7]. Cafeteria and traffic recordings come from DEMAND [8]. Gaussian white noise was generated with seed 5305. The noise is length-matched to each clean utterance, then scaled by the square root of clean power divided by the product of noise power and 10 raised to SNR/10. Adding the scaled noise gives the mixture. The complete 40 × 3 × 3 grid contains 360 mixtures. At 16 kHz, the 512-sample Hann window spans 32 ms and the 256-sample hop spans 16 ms.

3.2 Enhancement and reconstruction

3.2.1 Magnitude estimation

Spectral subtraction takes the maximum of the mixture magnitude minus mean noise magnitude and 0.02 times mixture magnitude. Wiener filtering first estimates speech power as mixture power minus mean noise power, clipped at zero. The gain is the estimated speech power divided by its sum with the noise power (plus a numerical stabiliser), with a minimum gain of 0.02. The mixture magnitude is multiplied by this gain.

3.2.2 Reconstruction conditions

Table 1. Reconstruction conditions

Condition family	Synthesis magnitude	Synthesis phase
Unprocessed reference	Mixture	Mixture
Spectral subtraction pair	Same subtraction estimate	Mixture or clean
Wiener pair	Same Wiener estimate	Mixture or clean
Exact-magnitude diagnostic	Clean reference	Mixture
MP-SENet pair	Same network estimate	Mixture or network

Classical reconstruction combines magnitude and phase by inverse STFT at the original signal length. Phase disagreement is the clean-magnitude-weighted mean absolute wrapped difference between the mixture and clean angles, in radians. Paired benefit is the clean-phase score minus the mixture-phase score for the same mixture and estimator.

3.3 Metrics and analysis

Signals are trimmed to a common length. The experiment uses standard STOI, 16 kHz wideband PESQ, and SI-SDR computed from mean-centred signals. The paired tables use one record per mixture and condition. Pooled Pearson r uses all mixtures for each estimator. Cell-centred r subtracts each noise-by-SNR cell mean from both the disagreement and the gain before correlating residuals. This descriptive check removes between-cell mean differences. The 95% percentile intervals use 10,000 bootstrap resamples of source utterances, retaining their mixtures together. They describe uncertainty within the observed two-speaker dataset.

The pretrained MP-SENet benchmark uses 167 mixtures, matched to classical scores by filename, noise and SNR. On the same inputs, the ablation fixes the estimated magnitude and swaps the synthesis phase using the model’s 400-sample Hann window, 100-sample hop and magnitude compression exponent 0.3. Scoring uses float waveforms after undoing input normalisation.

4 Results

4.1 Paired effects across 360 mixtures

Table 2. Paired phase-swap gains

Estimator	Score	Mean gain	95% interval	Positive	Negative
SS	STOI	+0.0170	0.0163 to 0.0178	360	0
SS	PESQ	+0.0661	0.0590 to 0.0728	354	6
SS	SI-SDR (dB)	+1.9602	1.7116 to 2.1446	349	11
Wiener	STOI	+0.0164	0.0156 to 0.0171	360	0
Wiener	PESQ	+0.0608	0.0533 to 0.0694	354	6
Wiener	SI-SDR (dB)	+1.8978	1.6251 to 2.0995	345	15

Table 2 gives the clean-phase score minus the noisy-phase score for 360 pairs per estimator, with no ties. Mean gains were positive for all three scores. STOI rose on every mixture. PESQ fell on six mixtures per estimator. SI-SDR fell on 11 spectral-subtraction mixtures and 15 Wiener mixtures.

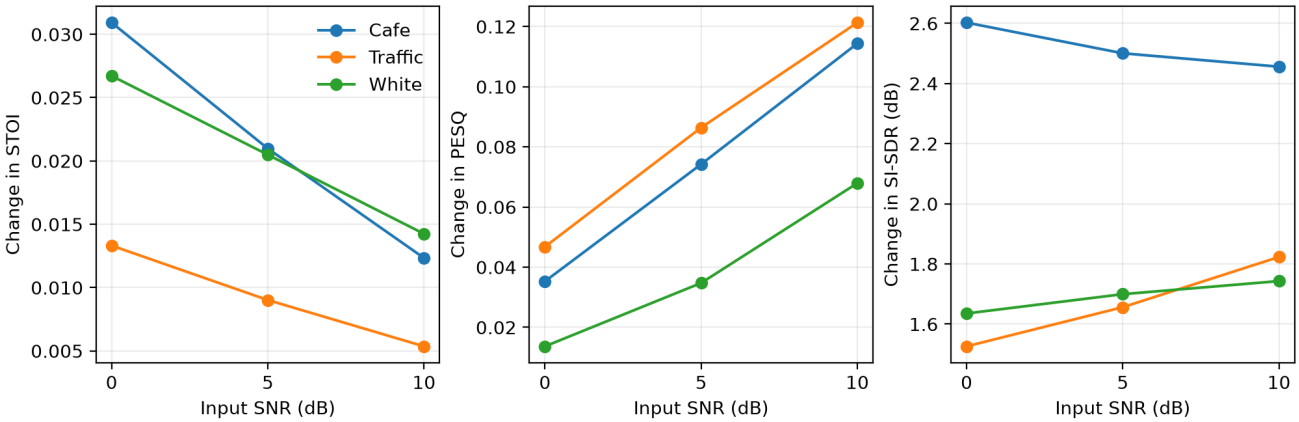


Figure 1. Spectral-subtraction gains by noise and SNR

Each point in Figure 1 averages 40 mixtures. For spectral subtraction in cafeteria noise, lowering input SNR from 10 to 0 dB increased mean STOI gain from 0.0123 to 0.0309. PESQ moved in the opposite direction, with gains of 0.1144 at 10 dB and 0.0353 at 0 dB. Mean SI-SDR gain was larger in cafeteria than in traffic or white noise at every tested SNR. The SNR trends for SI-SDR varied with noise type.

4.1.1 Comparison with the noisy input

Across the full grid, mean STOI rose from 0.8565 in the noisy mixture to 0.8593 after spectral subtraction with noisy phase and to 0.8763 after the phase swap. The first change comes from modifying magnitude. The second comes from modifying phase at the same estimated magnitude.

The largest mean angle disagreement was in white noise at 0 dB (0.508 rad). The largest spectral-subtraction STOI gain was in cafeteria noise at 0 dB (0.0309). Comparing these cells shows why the average angle alone cannot rank the benefit across all conditions.

4.2 Phase disagreement

A pooled correlation can reflect systematic differences across noise and SNR, variation among mixtures tested under the same conditions, or both. Comparing it with a cell-centred correlation helps separate these sources of association.

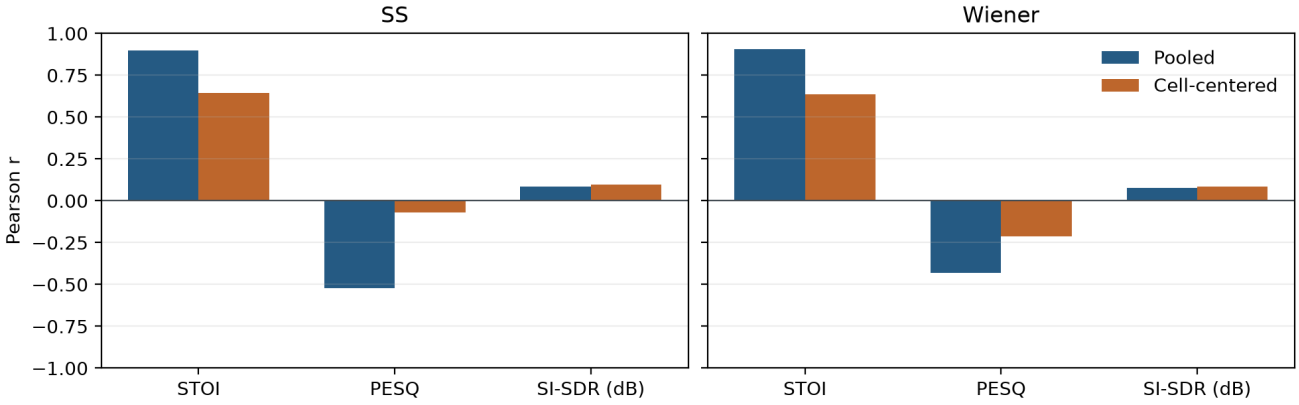


Figure 2. Pooled and cell-centred correlations

Figure 2 uses 360 observations per estimator. For spectral subtraction, STOI r fell from 0.897 to 0.643 after centring, and PESQ r moved from –0.523 to –0.068. Wiener STOI showed a similar residual association (0.635), while its PESQ association weakened from –0.429 to –0.213. SI-SDR r remained close to zero for both estimators despite a positive mean gain.

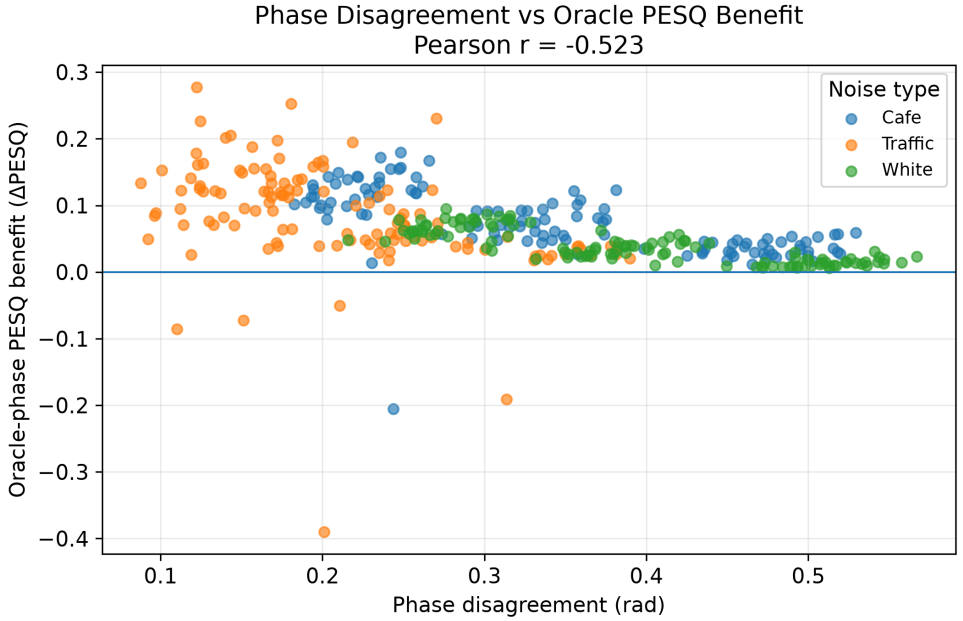


Figure 3. Phase disagreement and PESQ gain

Figure 3 pools the nine test cells and shows individual negative gains. Five of the six spectral-subtraction PESQ decreases occurred in traffic noise. The largest was –0.390 for p232_020 at 0 dB. The largest SI-SDR drop was –5.15 dB for a different utterance, p232_012 in traffic at 0 dB. These cases show that the two scores can flag different failures. Centring removes cell-level mean shifts but retains variation among utterances. For spectral subtraction, Figure 2 shows that the direction and strength of the STOI association differ from PESQ even after centring. This comparison does not remove other shared effects, such as speaker or utterance content. The angle also requires clean speech to compute, so it cannot directly select a phase-enhancement method from a noisy input alone.

4.3 Magnitude diagnostic and matched benchmark

4.3.1 Effect of supplying clean magnitude

Across all 360 mixtures, replacing the spectral-subtraction magnitude with the clean magnitude at the same noisy phase raised mean STOI by 0.1197, PESQ by 2.1244 and SI-SDR by 6.08 dB. These gains exceeded the corresponding phase-swap means in Table 2. The comparison identifies substantial headroom in the estimated magnitude under this reconstruction setup. Both diagnostic changes use clean reference information.

4.3.2 Matched MP-SENet comparison

Table 3. Mean scores on matched mixtures

Condition	STOI	PESQ	SI-SDR (dB)
Noisy	0.8595	1.2844	5.03
SS + noisy phase	0.8597	1.4639	9.16
SS + clean oracle phase	0.8749	1.5312	11.09
Wiener + noisy phase	0.8573	1.4098	8.53
Wiener + clean oracle phase	0.8720	1.4765	10.40
Perfect magnitude + noisy phase	0.9794	3.6227	15.13
MP-SENet	0.9231	2.3356	14.58

Table 3 compares the same 167 mixtures from 27 utterances. Exact magnitude and clean phase use the clean reference. Against spectral subtraction with clean phase, MP-SENet gained 0.8044 PESQ (95% interval 0.7636 to 0.8510) and improved PESQ on 167/167 mixtures. It improved STOI on 161/167 and SI-SDR on 141/167. Against the exact-magnitude diagnostic, its mean SI-SDR was 0.55 dB lower even though it scored higher on 85/167 mixtures. The mean and the count therefore describe different properties of the paired differences.

4.4 Cafeteria example at 0 dB

Table 4. Scores for p232_001 in cafeteria noise

Condition	STOI	PESQ	SI-SDR (dB)
Noisy	0.7343	1.1941	-0.13
SS + noisy phase	0.7277	1.2226	2.23
SS + clean oracle phase	0.7512	1.2703	4.75
Wiener + noisy phase	0.7273	1.1957	2.15
Wiener + clean oracle phase	0.7481	1.2421	4.60
MP-SENet	0.8403	1.9166	11.03

Table 4 uses the representative audio evaluation. Spectral subtraction raised SI-SDR from −0.13 to 2.23 dB with noisy phase while STOI fell from 0.7343 to 0.7277. Substituting clean phase raised STOI to 0.7512 and SI-SDR to 4.75 dB. The example demonstrates how improved waveform fidelity can coexist with an intelligibility proxy that first worsens and then improves. MP-SENet gave 0.8403 STOI and 11.03 dB SI-SDR on this example.

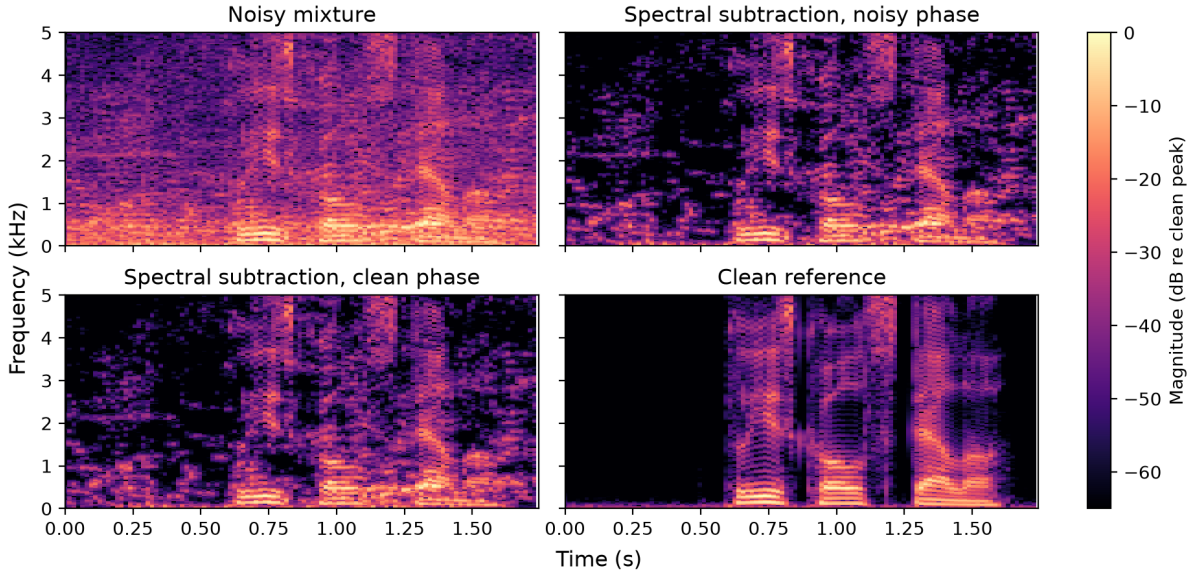


Figure 4. Spectrograms of the cafeteria example

Figure 4 uses one colour scale referenced to the clean peak. Before speech begins at about 0.6 s, the noisy mixture has energy across much of the displayed band. Both spectral-subtraction outputs attenuate that energy, but residual energy remains where the clean reference is near silence. The two enhanced panels are visually similar in several speech regions even though the phase swap changes the waveform and the scores in Table 4. These are magnitude spectrograms of the reconstructed audio. They illustrate residual noise and speech structure, while the waveform scores quantify the effects of the phase substitution.

Listen to the aligned audio example.

4.5 MP-SENet phase ablation

Table 5. Paired MP-SENet phase results

Metric	Mixture phase	Model phase	Mean gain (95% interval)	Improved
STOI	0.9192	0.9231	+0.0040 (+0.0034, +0.0045)	149/167
PESQ	2.2182	2.3358	+0.1176 (+0.1054, +0.1319)	120/167
SI-SDR (dB)	12.99	14.58	+1.59 (+1.48, +1.69)	166/167

On the 167 matched mixtures from 27 utterances, mean scores improved on all three metrics. Table 5 reports model phase minus mixture phase. The intervals resample utterances, retaining all matched mixtures belonging to each sampled utterance. The model-phase scores agreed with the archived PCM16 scores to within 0.00034 STOI, 0.00290 PESQ and 0.0094 dB SI-SDR.

Table 6. Mean phase gains by condition

Noise	Input SNR (dB)	N	STOI gain	PESQ gain	SI-SDR gain (dB)
Cafe	0	19	+0.0134	+0.1206	+2.41
Cafe	5	19	+0.0070	+0.1767	+2.05
Cafe	10	19	+0.0036	+0.1722	+1.48
Traffic	0	20	+0.0046	+0.2050	+2.57
Traffic	5	18	+0.0025	+0.2161	+2.72
Traffic	10	21	+0.0011	+0.1588	+1.86
White	0	17	+0.0018	-0.0083	+0.37
White	5	17	+0.0005	-0.0117	+0.24
White	10	17	+0.0002	-0.0157	+0.14

Mean STOI and SI-SDR gains were positive in all nine cells. In cafeteria and traffic noise, mean PESQ gains ranged from 0.121 to 0.216. In white noise, mean PESQ changed by −0.008 to −0.016 while mean SI-SDR still rose by 0.14 to 0.37 dB. The white-noise PESQ changes are small and describe the tested mixtures. A listening test would be needed to judge their perceptual importance.

The largest PESQ drop was 0.055 for p232_010 in white noise at 10 dB. The sole SI-SDR decline was 0.006 dB for p232_001 under the same condition. Different utterances produced these exceptions, reinforcing the need to report each metric separately.

5 Discussion

5.1 Why the metrics differ

5.1.1 STOI and input SNR

In a bin where the interference vector is comparable to the speech vector, adding the two can rotate the resulting coefficient substantially. This is more likely at low SNR, but global SNR alone does not determine the local interference-to-speech ratio. Changing phase can alter reconstructed envelopes as inverse STFT combines overlapping frames. The within-cell association ($r = 0.643$) suggests that phase disagreement tracks some of this variation even among utterances at the same noise type and input SNR. Locating the responsible bins or estimating listener intelligibility would require a time-frequency ablation or listening experiment.

5.1.2 PESQ across test conditions

At higher SNR, the classical magnitude estimate may contain less noise-related error, leaving more speech structure for the phase substitution to preserve. At lower SNR, residual interference or excessive attenuation in that magnitude may constrain the PESQ gain. This mechanism is consistent with the cafeteria result and the large exact-magnitude gains. This explanation remains a hypothesis. For spectral subtraction, the correlation moved from -0.523 to -0.068 after centring. The pooled trend therefore provides little basis for predicting which utterances will gain more PESQ within a given noise and SNR condition.

5.1.3 SI-SDR and phase disagreement

SI-SDR compares waveform residual energy after scale alignment [11]. Removing a phase error can improve alignment, which is consistent with the mean gains near 2 dB. The mean absolute angle weights frequency bins by clean magnitude and loses their timing and pattern. It need not track the change in residual energy. Individual negative gains are possible when the imperfect magnitude and the new phase interact during overlap-add, or when noise removal and speech distortion trade off differently under a given metric.

5.2 Interpreting the matched benchmark

MP-SENet beat clean-phase spectral subtraction on PESQ in all 167 mixtures. The phase-only swap improved PESQ in 120. The full-system comparison also changes the magnitude and the estimator. The ablation measures the synthesis-phase effect conditional on the model magnitude. Under white noise, switching to the estimated angle reduced mean PESQ at every SNR even as SI-SDR rose. SI-SDR rewards waveform alignment, while PESQ responds to perceptually weighted spectral changes. These trends establish a metric trade-off for this sample. Lu et al. [6] report PESQ 3.50 on their standard benchmark. This study's 2.34 describes the smaller custom mixture grid.

5.3 Relation to prior work

The phase-swap results are consistent with Paliwal et al. [3] on the relevance of phase. This project extends that comparison to two fixed classical magnitude estimates across nine noise and SNR cells. The clean-magnitude substitution produced larger mean gains on all three metrics. Both diagnostics start from the same spectral-subtraction baseline, and each changes one component using the clean reference. This makes magnitude estimation a priority for a follow-up under these conditions. A joint substitution would be needed to test whether the two effects interact. The results support the motivation for joint estimation in MP-SENet [12]. Runtime and encoder modifications [13] remain open questions.

6 Conclusion

6.1 Main findings

For the tested classical estimators, noisy-phase reuse limited mean performance by roughly 0.016–0.017 STOI, 0.061–0.066 PESQ and 1.90–1.96 dB SI-SDR relative to supplying clean phase at the same estimated magnitude. The mean STOI gain was largest at 0 dB within each noise type and remained associated with phase disagreement after cell centring. The analogous PESQ relationship was substantially weaker after centring. Thus the experiment answers the research question for this grid and this reconstruction configuration. The magnitude diagnostic locates additional headroom. Within MP-SENet, using the estimated phase increased SI-SDR in 166 of 167 matched mixtures, with a less consistent benefit for PESQ.

6.2 Limitations and follow-up experiments

Only two speakers and two recorded environments were evaluated. Repeated mixtures from each utterance share speech content, and each recorded noise represents one environment sample. These observations describe the archive and do not establish population-level effects. A repeat with more held-out speakers, independent noise excerpts and additional window durations would test how stable the metric trends are. The utterance-level bootstrap retains this dependence across repeated mixtures. The intervals remain conditional on the observed speakers and recording setup.

The MP-SENet subset covers 167 of 360 mixtures and was determined by completion, with 18 unique utterances from p232 and nine from p257. Café, traffic and white noise account for 57, 59 and 51 mixtures. The 0, 5 and 10 dB counts are 56, 54 and 57. The matched and excluded sets already differ in mean noisy PESQ (1.2844 vs 1.1918) and spectral-subtraction SI-SDR (9.16 vs 9.86 dB). Matching controls which examples enter the comparison, while this difference limits generalisation to the entire grid. Completing inference on the full grid would resolve that coverage gap.

The listening example illustrates one outcome and no formal listening study was conducted. A level-matched, blinded comparison could test whether the metric gains correspond to perceived clarity or quality. The phase ablation measures the effect of substituting the angle at synthesis. The network still receives mixture phase and produces both outputs jointly. Retraining with the phase path removed would test a different question about the architecture.

6.3 What I would change in a repeat

I would keep the paired phase swap and fix the matched input list in advance. I would examine the negative-gain traffic cases with shared-scale spectrograms and listening, and repeat the fixed-magnitude phase ablation on a broader matched set.

6.4 Reproducing the results

Run `python experiments/demo.py` with the bundled VoiceBank and DEMAND WAV files to generate audio and a score table at the three SNRs. Run `python experiments/analyze_final_results.py` to regenerate the paired tables, subgroup results and figures from the archived CSVs. The data and model instructions record the verified noise channels, model revision and checkpoint checksums. The accompanying phase experiment archive provides the ablation code, 167 paired scores and scripts for both summary tables.

The corpus, noise recordings, pretrained model and metric implementations are credited in the references and repository. AI assistance was used to help revise the analysis code, prepare the report and assemble the demonstration. The demonstration uses English captions and speech examples.

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Code resources

MP-SENet · pystoi · PESQ Python wrapper